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**An Efficient and Interpretable Framework for Social Media Sentiment
Classification Using Traditional Machine Learning Techniques**

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Abstract

The study explores the opportunity to use traditional machine learning algorithms to assess sentiment conveyed through casual digital communication, with a focus on the Sentiment140 dataset. A tailored text-cleaning and normalisation strategy was employed to address the variability of social media language, including emoji handling, negations, and lexical inconsistencies. To numerically represent text, term weighting was implemented using TF-IDF, capturing both isolated words and short phrase combinations. The sentiment classification task involved training and comparing five supervised algorithms: Logistic Regression, Linear Support Vector Classifier, Stochastic Gradient Descent, Bernoulli, and Multinomial Naive Bayes. Among them, Logistic Regression demonstrated the most balanced and consistent results across standard evaluation metrics. Supplementary analysis of word frequency patterns helped reveal sentiment trends, visually presented through word clouds. Reliability was confirmed via cross-validation and confusion matrix inspection, supporting the scalability of the framework for sentiment modeling in digital environments.

Chapter 1: Introduction

1.1 Study Background

The accelerated development of platforms like Twitter, Facebook, and Reddit has significantly changed the way individuals communicate, share ideas, and engage in online discussions. These sites continuously produce large volumes of text-based, unstructured content that offers researchers and organisations valuable insights into social trends, consumer sentiment, and behavioural patterns (Kandasamy & Murugasamy, 2021). Yet, the casual, brief, and ever-evolving nature of social media content presents major obstacles to manual evaluation.

Techniques in Natural Language Processing (NLP) and machine learning have emerged as effective solutions for automating sentiment identification, allowing for scalable and systematic analyses of public opinion. These tools aid in determining emotional tone, monitoring customer feedback, and anticipating public reactions to events or policy. Consequently, sentiment analysis has developed into a key application of NLP, offering tools to uncover meaningful insights from large-scale social media data (Kaur & Sharma, 2023).

1.2 Problem Statement

Despite significant development in sentiment analysis, many models are geared towards grammatically structured and formal texts like product reviews or news stories. In contrast, social media text is frequently informal, using slang, emoticons, abbreviations, and even sarcasm (Wankhade et al., 2022). These characteristics undermine the effectiveness of conventional sentiment analysis techniques, often causing misinterpretations and lower prediction accuracy.

Although advanced architectures such as LSTM networks and BERT enhance context understanding, they are computationally demanding and difficult to interpret (Ghimire et al., 2022). Their opaque decision-making procedures can be problematic in areas where transparency and explanation are required. Therefore, developing a sentiment analysis approach that balances effectiveness, speed, and clarity is essential.

1.3 Research Aim

This study mainly seeks to develop a system that is both scalable and easy to interpret for analysing sentiments in social media content with classical machine learning approaches. The objective is to achieve high classification accuracy while maintaining low computational requirements and promoting model transparency.

1.4 Research Objectives

The following detailed goals guide the direction of this research:

- Applying NLP-based preprocessing strategies that include tokenisation, lemmatisation, removal of stopwords, emoji interpretation, and handling of negation, to collect and clean social media content.
- Implementing and assessing sentiment classification models using models like Logistic Regression, Naive Bayes, Stochastic Gradient Descent, and Support Vector Machines (SVM).
- Investigating sentiment-bearing trends using frequency visualisations and keyword analysis to gain thematic insights.
- Increasing interpretability by visualising sentiment classification findings with tools like word clouds, bar charts, and confusion matrices.
- Improving the models iteratively by evaluating and contrasting model outputs based on metrics such as recall, precision, accuracy, and the F1-score.

1.5 Research Questions

Aligning with the objectives outlined earlier, this study seeks to explore the questions below:

- To what extent are machine learning and NLP methods capable of identifying sentiment in informal social media content?
- What are the drawbacks and strengths of using traditional machine learning approaches for sentiment classification?
- What kinds of themes and sentiment trends may be identified from social media content?
- How might sentiment analysis results be better understood and communicated via data visualisations?

- What techniques can enhance sentiment analysis frameworks' scalability, effectiveness, and usability?

1.6 Study Scope and Limitations

This study looks specifically at English-language text collected from social media networks. It excludes non-textual elements, such as photos, music, and video. Furthermore, real-time sentiment tracking is outside the scope of this work. While deep learning techniques are extensively explored in previous research, their application is frequently confined to classic machine learning algorithms due to their efficiency, ease of use, and interpretability. All analyses are based on publicly available datasets.

1.7 Significance of Study

By demonstrating the strength of classic machine learning algorithms integrated with optimised NLP preparation, this study enhances easily accessible sentiment analysis tools. For academics and organisations with limited computational resources, the suggested framework provides a workable solution that allows for large-scale public opinion analysis, trend identification, and sentiment tracking (Rodríguez-Ibáñez et al., 2023). By focusing on the challenges presented by informal and loud language on social media, the research also fills a significant gap in the existing body of work. The findings have greater implications for applied NLP model design, especially in cases in which efficiency and interpretability are essential.

1.8 Summary

The research adopts an organised approach based on existing data science techniques. It starts with data collection from pre-labelled Twitter datasets and ends with extensive text preparation using conventional NLP approaches (Abdul-Rahman et al., 2021). Several machine learning classifiers are developed and tested using statistical performance indicators. Data visualisation techniques, including visual tools like confusion matrices and word cloud representations, are used to effectively explain findings. Although sophisticated deep learning techniques are mentioned, the focus is on developing a strong yet interpretable pipeline with lightweight models.

Chapter 2: Literature Review

2.1 Introduction

This section analyses recent studies on how sentiment is analysed, focusing particularly on methods applied to examine social media content utilising NLP (Natural Language Processing) and standard machine learning techniques. It adds a theoretical foundation to the study, assesses various approaches, identifies gaps in current research, and situates the present investigation within that framework. The literature is organised topically to facilitate clarity and show the progress of sentiment analysis methodologies.

2.2 Theoretical Framework

2.2.1 Linguistic Appraisal Theory

Linguistic Appraisal Theory addresses how emotional attitudes are conveyed through language, including adjectives, intensifiers, and negations (Mitra Mirzaaghabeyk, 2022). This theory promotes the extraction of sentiment-bearing characteristics and justifies the use of syntactic and grammatical aspects in text preparation. However, it has limitations in its ability to deal with emoticons and multimodal inputs, which are common in social media.

2.2.2 Context Integration Theory

This approach emphasises the layered character of emotion grounded in micro (words), meso (topic), and macro (cultural or historical) settings (Muhammad et al., 2016). Understanding the context is critical for appropriate sentiment analysis, particularly on networks like Twitter. Unfortunately, many models simply include word-level characteristics, omitting more general indications like user intent or event-specific meaning.

2.2.3 Multimodal Communication Theory

According to Yus (2021), hashtags, photos, and emoticons enhance written language and affect sentiment. The hypothesis guides the decision to use emoji translation during preprocessing to preserve emotional complexity in tweets, even if this project is restricted to text.

2.3 Lexicon-Based Approaches

Lexicon-based methods such as VADER and TextBlob assign sentiment scores based on word polarity dictionaries. VADER, designed for social media, performs well on simple texts due to its rule-based handling of punctuation and capitalisation (Ogunleye et al., 2024). However, these models lack adaptability and struggle with sarcasm or informal language, leading to misclassifications in short, ambiguous posts. Despite being simple to use, TextBlob frequently ignores negation and emotion unusual to a certain domain. These drawbacks support the choice to confine this study to using lexicon-based approaches as theoretical foundations.

2.4 Machine Learning Methods

2.4.1 Naive Bayes and Logistic Regression

Naive Bayes (NB) and Logistic Regression (LR) are commonly used because of their straightforward nature and reliable performance with sparse, high-dimensional datasets. NB assumes feature independence, which simplifies computation but may reduce accuracy with correlated features (Nikolić et al., 2020). LR, in contrast, is more robust to such dependencies and performs well with TF-IDF representations. Both models are suitable for text-based sentiment classification tasks due to their ease of interpretation and scalability, which makes them ideal for this project's objective of developing an efficient, explainable model.

2.4.2 Support Vector Machines and Stochastic Gradient Descent

The ability to handle non-linear boundaries using kernel functions makes Support Vector Machines (SVMs) effective. They consistently deliver high accuracy across sentiment datasets but can be computationally expensive during training (Xu et al., 2022). Stochastic Gradient Descent (SGD), by contrast, is efficient for large-scale text classification and is particularly effective when combined with regularisation. Both models are included in this study due to their established effectiveness and success in analysing social media sentiment and their compatibility with TF-IDF features.

2.5 Deep Learning and Transformer Models

Deep learning approaches are not implemented in this specific study directly, but they still have relevance in the broader academic landscape. Long Short-Term Memory (LSTM) networks are

particularly useful in detecting consecutive patterns in data and are especially helpful for analysing complex sentence patterns (Pimpalkar & Raj, 2022).

Transformer-based models, for example, BERT, understand word context in both directions and frequently outperform many traditional methods on benchmark datasets (Shreyashree et al., 2022). However, models like these often depend on extensive computational power and large-scale annotated datasets, making them less suitable for use in settings with limited resources or time constraints. Their complex decision-making process also contradicts this study's goal of using interpretable models.

2.6 Challenges in Social Media Sentiment Analysis

Analysing user sentiment on social media platforms such as Twitter presents several unique difficulties:

- Spelling mistakes, abbreviations, and slang might reduce accuracy.
- Difficult to detect without contextual or user-level data (Boutsikaris & Polykalas, 2024).
- Tweets may not provide sufficient information for standard models due to their short text length.
- The presence of multiple languages within a dataset adds complexity to the classification process (Agüero-Torales et al., 2021).
- The models might inherit biases from training data, raising ethical questions about data usage and permission (Kaddoura & Al Hussein, 2023).

These issues reinforce the need for robust preprocessing, transparent algorithms, and ethical consideration.

2.7 Application of Sentiment Analysis

In a variety of fields, sentiment analysis contributes significantly. Such as:

- Business: Companies use it to monitor brand sentiment and enhance customer service (Ayoolu et al., 2024).
- Politics: It helps gauge public opinion on candidates and policies (Verma, 2022).
- Mental Health: Social media sentiment has been linked to detecting emotional distress in users (Kim et al., 2023).

- Crisis response: Governments monitor reactions to emergencies or pandemics.

These examples demonstrate the practical benefits of developing efficient, adaptive sentiment models.

2.8 Emerging Trends

Recent trends include:

- Aspect-Based Sentiment Analysis (ABSA): This is a technique that examines user opinions targeted at specific aspects or features within a topic (Maitama et al., 2020).
- Multimodal Analysis: Integrates text, images, and audio (Gandhi et al., 2023).
- Explainable AI (XAI): Increasing focus on models that can explain their decisions (Ainura Tursunaliyeva et al., 2024).
- Real-Time Analysis: Systems designed to process and visualise sentiment from live data streams (Mehmood & Anees, 2020).

While some paths show potential, many still struggle with scalability and interpretability.

2.9 Literature Gaps and Relevance to This Study

Significant gaps still exist despite advancements such as:

- Few research studies create lightweight models that are tailored for brief, noisy social media material.
- Existing tools often lack transparency, which limits their use in decision-making.
- Real-world constraints (e.g., time, data privacy, resources) are rarely considered in model design.
- Sarcasm detection and domain adaptation remain weak in most models.

By adopting interpretable, resource-efficient machine learning models and concentrating on preprocessing methods and assessment procedures appropriate for social media settings, this work tackles these problems.

2.10 Summary Table: Comparative Overview of Methods

Study/Method	Technique	Strengths	Limitations	Relevance to This Study
VADER	Lexicon-based	Simple, social media-tuned	Struggles with sarcasm and context	Reviewed as a theoretical baseline
TextBlob	Lexicon-based	Easy to implement	Lacks negation handling	Literature reference only
Naïve Bayes	Machine Learning	Fast, scalable	Assumes feature independence	Implemented and evaluated
Logistic Regression	Machine Learning	Interpretable, good with TF-IDF	May underperform on non-linear patterns	Implemented and evaluated
Support Vector Machine	Machine Learning	High accuracy, strong classification	Training can be computationally expensive	Implemented and evaluated
Stochastic Gradient Descent	Machine Learning	Scalable, handles large datasets	Sensitive to hyperparameters	Implemented and evaluated
BERT / LSTM	Deep Learning / Transformers	Context-aware, strong accuracy	High computational cost, less transparent	Acknowledged, not implemented

2.11 Summary

With an emphasis on social media environments, this chapter examined the theoretical and applied landscape for analysing sentiment. It reviewed the weaknesses and strengths of machine learning approaches alongside lexicon-driven techniques, noting their limitations in areas such as interpretability, complexity, and overall accuracy, even with widespread adoption. The necessity for well-rounded, resource-conscious solutions is further highlighted by the shortcomings of transformer models and deep learning. This review supports the study's use of extensive preprocessing techniques and conventional machine learning models by pointing out current issues and gaps. How these realisations influenced the research technique is described in the next chapter.

Chapter 3: Methodology

3.1 Introduction to Research Methodology

This section outlines the process of creating a sentiment classification model that is both user-friendly and reliable, using NLP-based techniques. This study applies traditional machine learning methods in a structured, task-driven manner to develop a sentiment classification model that is both computationally efficient and able to accurately detect sentiment-driven trends in social media content. First, the tool collects data from a pre-labelled Twitter set, prepares the data and the text, and then applies the sentiment models such as Logistic Regression, Linear SVC, Multinomial Naive Bayes, Bernoulli Naive Bayes, and Stochastic Gradient Descent Classifier. Performance measurement relies on precision, recall, accuracy, and F1-score, and using visualisations highlights the words that are often related to various sentiments (Ainura Tursunaliyeva et al., 2024). This happens because the tools used to visualise the data allow the insights to be presented more clearly. Additionally, the trained model is prepared for potential deployment within a real-time sentiment analysis interface, allowing sentiment predictions to be generated dynamically from user input.

3.1.1 Methodological Framework: CRISP-DM

This methodological approach adopts the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which is a well-established model for structuring data science and machine learning projects. CRISP-DM outlines a flexible and iterative procedure that follows six key stages: understanding the business context, examining the data, preparing the data, evaluating outcomes, and deployment.

This framework was selected due to its alignment with the exploratory and iterative nature of machine learning workflows, enabling continuous refinement and feedback across all project stages. In the context of this study, the framework guided the end-to-end sentiment analysis process as follows:

- **Business Understanding:** Defined the objective of classifying sentiment in Twitter posts to assess public opinion patterns.

- **Data Understanding:** Involved analysing the structure and distribution of the Sentiment140 dataset to identify challenges specific to social media language.
- **Data Preparation:** All the necessary steps for text preprocessing were applied to clean and normalise the text, including emoji translation, stopword removal, character normalisation, negation handling, and lemmatisation.
- **Modeling:** Implemented multiple supervised learning models using TF-IDF features, including Logistic Regression, Naive Bayes variants, SVC, and the SGD model.
- **Evaluation:** The model's performance was assessed using a diverse set of evaluation metrics, namely precision, recall, accuracy, and F1-score, along with confusion matrices to gain a deeper understanding of its effectiveness.
- **Deployment:** Exported trained models and the vectoriser using serialisation tools for real-time prediction applications.

The CRISP-DM framework not only provides a logical flow for the research process but also supports iterative refinement, allowing insights from evaluation to inform modifications in data preparation and model tuning. This structured yet adaptable approach enhances the reliability, transparency, and reproducibility of the overall methodology.

3.2 Research Philosophy and Approach

This research is grounded in a pragmatic philosophical stance, which prioritises using effective and applicable methods based on the nature of the research question, rather than strictly aligning with a single philosophical thought (Allemang et al., 2021). Pragmatism is especially appropriate for sentiment analysis, where textual data often carries both quantitative properties (e.g., polarity scores) and qualitative subtleties (e.g., sarcasm, cultural references).

The research design integrates both deductive and inductive reasoning strategies. Deductive logic informs the application of established algorithms and preprocessing techniques, while inductive analysis is used to derive insights from model performance and error patterns, facilitating iterative refinement. Widely accepted NLP techniques and models are used deductively to process the data, while model performance and error trends are explored inductively to guide ongoing adjustments.

This combination ensures that theoretical knowledge is applied while allowing data-driven insights to inform methodological adjustments.

The methodological workflow is applied and iterative, aligned to improve sentiment classification performance in a dynamic environment. Adjustments are frequently made to accommodate the informal language styles found on social platforms, which often involve casual expressions, negations, and slang. This approach is flexible and ensures the system remains responsive and capable of ongoing refinement.

3.3 Data Collection Strategy

This study makes use of the Sentiment140 dataset, a publicly available corpus containing tweets (about 1.6 million) pre-labelled as either having positive or negative sentiment (Pathak et al., 2021). This dataset is well-established in sentiment analysis research due to its large volume, real-world context, and pre-labelled structure, which supports efficient supervised learning without requiring additional annotation.

The dataset used in this study is publicly accessible via Kaggle:

<https://www.kaggle.com/datasets/kazanova/sentiment140>

To prepare the dataset for developing the model, a 95% training and 5% testing split was employed. Although a 70-15-15 split is commonly used in academic studies, the substantial volume of data in the Sentiment140 dataset permits a more aggressive training allocation. This study adopts a 95% training and 5% testing split, ensuring sufficient exposure to diverse sentiment expressions while maintaining statistical validity in evaluation. With approximately 80,000 samples remaining for testing, statistical reliability is maintained while maximising model exposure to diverse sentiment expressions during training.

Instead of a separate validation set, the training process uses cross-validation techniques to monitor generalisation performance. This approach reduces computational overhead and aligns with real-world practices where deployment efficiency is critical (Kumar et al., 2025).

Importantly, the dataset contains no personal identifiable information, and all metadata, such as usernames and timestamps, are either anonymised or excluded. As such, its use complies with ethical standards for data privacy and responsible research conduct.

3.4 Data Preprocessing Techniques

For sentiment analysis, preparing text data is a crucial part as it transforms unstructured content into a format that can be effectively processed by machine learning models. This is particularly important to handle the informal and inconsistent language that is often found on platforms like Twitter.

As part of the data preparation process, only the columns essential for the sentiment classification task were retained. Specifically, the dataset was reduced to two columns: sentiment, which represents the label, and text, which contains the actual tweet content. All other metadata fields, such as tweet IDs, usernames, timestamps, and query tags, were excluded to simplify the input and keep the focus on the sentiment expressed in the tweet itself.

To manage issues such as noise, slang, abbreviations, and special symbols, a custom preprocessing sequence was implemented for this project. Implemented in Python using NLP libraries like NLTK, Regex, and custom functions, the modular pipeline is optimised for social media text and scalable for large datasets.

Step 1- Text Cleaning: This step includes:

- Converting to lowercase to ensure uniformity (e.g., 'Good' → 'good').
- Removing URLs, replacing links with a placeholder token 'URL' to retain context without adding noise.
- Handling user mentions by replacing usernames with 'USER' to anonymise content while maintaining context.
- Removing special characters, stripping out punctuation and non-alphanumeric symbols, except for important ones like hashtags.
- Reducing repeated characters, limiting sequences of more than two repeated letters (e.g., 'loooove' → 'loove').

These preprocessing steps are essential for reducing linguistic noise while preserving features tied to emotion. This ultimately enhances the quality and clarity of input data before it enters machine learning pipelines.

Step 2- Emoji Translation: Emojis are highly expressive in social media sentiment. Common emojis are translated into textual descriptions (e.g., 😊 → 'smiling face with smiling eyes') to retain emotional cues often lost in basic text cleaning (Kumar et al., 2025).

Step 3- Stop Word Removal: A custom stop word list is used to remove common but sentiment-neutral words (e.g., 'the', 'is', 'in') while retaining words like 'not' and 'very' that influence sentiment polarity. This improves the model's ability to isolate emotionally charged or polarising expressions (Kumar et al., 2024).

Step 4- Tokenisation: The cleaned text is broken down into individual units like words or symbols, accounting for social media language elements such as hashtags, abbreviations, and emoticons. This enables the model to process and analyse each sentiment-bearing unit effectively (Yadav et al., 2024).

Step 5- Normalisation: This step includes expanding short forms (e.g., "don't" → "do not") and correcting common spelling issues based on context. These practices help standardise social media text, which often contains informal expressions and typos (ÇetinV & Yildiz, 2022). Future iterations of the preprocessing pipeline may incorporate automated spelling correction to further standardise informal text and reduce classification ambiguity.

Step 6- Lemmatisation: It reduces words to their base forms while preserving grammatical context (e.g., 'running' → 'run'). This is preferred over stemming to maintain semantic clarity in sentiment detection (Qiang et al., 2024).

Step 7- Negation Handling: Negation is addressed by appending a suffix '_NEG' to words following negation terms (e.g., 'not good' → 'not_good'), improving the model's understanding of sentiment shifts (Yadav et al., 2024).

Step 8- Final Preprocessed Output: The final output is a clean, tokenised, and sentiment-aware version of the original tweet text. It preserves critical indicators like emojis, negations, and linguistic cues while removing irrelevant noise, thereby enabling effective model training and learning.

3.5 Model Development and Implementation

This section outlines the machine learning classifiers employed for analysing sentiment in social media content. A supervised learning framework was used, where algorithms learn from annotated training data to determine sentiment polarity. Python and the scikit-learn library were used to build the models. The five classification algorithms were chosen based on their consistent performance in processing high-dimensional text data, computational efficiency with broad use in sentiment analysis literature (Kumar et al., 2024; Pal & Bhushan, 2024):

1. Bernoulli Naive Bayes (BNB)
2. Multinomial Naive Bayes (MNB)
3. Logistic Regression (LR)
4. Linear Support Vector Classifier (SVC)
5. Stochastic Gradient Descent Classifier (SGD)

The selected models represent a diverse range of algorithmic paradigms, including probabilistic classifiers (Naive Bayes variants), linear models (Logistic Regression, Linear SVC), and iterative optimisers (Stochastic Gradient Descent), allowing for comprehensive performance evaluation across varied methodological approaches.

Feature Extraction Using TF-IDF:

Term Frequency-Inverse Document Frequency (TF-IDF) vectoriser was applied to transform the cleaned dataset into a machine-readable numeric format. This method emphasises terms that are statistically significant within a document but less common across the full dataset, helping to prioritise sentiment-relevant features while downplaying high-frequency, low-value words (Huang et al., 2021).

The vectoriser was configured as follows:

`ngram_range = (1, 2)`: It captures single words (unigrams) and two-word combinations (bigrams).

`max_features = 500,000`: Limits the feature space for computational efficiency.

The vectoriser was fine-tuned using `max_df = 0.9`, `min_df = 5`, and `sublinear_tf = True` to exclude uninformative extremes and apply logarithmic term frequency scaling, in line with the sparse nature of tweet data. This configuration allows for contextual learning without overwhelming

model complexity. After fitting on the training data, the vectoriser was used to transform both training and testing text sets.

Model Training Process: Each of the five classifiers was trained on approximately 1.5 million tweets (95% of the Sentiment140 dataset). During training, the models learned to associate TF-IDF features with sentiment labels, optimising internal parameters to reduce classification error. The remaining 5% of the data (about 80,000 tweets) was reserved for testing.

Using a large training set ensured broad exposure to diverse linguistic structures, improving the model's generalisation capabilities. This helps mitigate underfitting, especially for models like Linear SVC and SGD, which perform well in high-dimensional vector spaces. For Logistic Regression and Linear SVC, hyperparameters were optimised using "GridSearchCV" with five-fold cross-validation, enabling systematic selection of the most effective regularisation strength (C).

Model Deployment: Upon completion of training, each model, along with the corresponding TF-IDF vectoriser, was serialised using Python's "pickle" library. This facilitates rapid loading for downstream applications and supports efficient deployment in real-time sentiment prediction contexts. Among the tested algorithms, Logistic Regression was chosen for live implementation because it provided an optimal trade-off between computational efficiency and classification accuracy.

3.6 Evaluation Framework

This section outlines the evaluation strategy used to examine the performance and reliability of the sentiment analysis models. Assessing how well the models identify sentiment categories across various social media inputs is the primary aim of this process.

Evaluation Metrics:

Four commonly used performance indicators were applied:

- **Precision:** Indicates the percentage of predicted positives that are correct.
- **Recall:** Reflects the proportion of actual positive cases that were successfully identified.

- **Accuracy:** Measures the ratio of correct predictions to the total number of predictions made.
- **F1-Score:** The harmonic average of precision and recall that balances both metrics to offer a more complete performance view.

Together, these metrics provide insight into the correctness, completeness, and robustness of each classifier's predictions (Naidu et al., 2023).

Confusion Matrix Analysis:

A confusion matrix was generated for each classifier to visualise the distribution of correctly and incorrectly predicted outcomes, which includes True Negatives (TN), True Positives (TP), False Negatives (FN), and False Positives (FP). The matrix enables detailed error analysis, highlighting specific areas where the model may have underperformed or overgeneralized across sentiment classes.

Cross-Validation Strategy:

To further validate the results, a five-fold cross-validation method was applied during training. The dataset, in this method, is divided into five segments, rotating through them such that each segment serves once as a validation set while the other four are used for training. This ensures that each data point is tested, increasing generalisability and reducing overfitting (Khan et al., 2025).

The evaluation framework was designed not only to assess final model performance but also to guide iterative improvements during model development.

3.7 Topic Modelling and Theme Identification

In addition to sentiment classification, the study also includes an analysis of prominent terms to uncover thematic patterns in the dataset. Instead of using probabilistic models like Latent Dirichlet Allocation (LDA), the study relies on TF-IDF-based keyword extraction to highlight terms most representative of each sentiment class.

This approach helps identify which words or phrases are most frequently associated with positive or negative sentiments. By analysing these high-weighted terms, it becomes possible to detect

recurring topics or themes such as emotional reactions, product feedback, or public opinion, embedded in social media discourse (Huang et al., 2021).

While this method does not generate formal topics in the probabilistic sense, it offers valuable insight into the language used within each sentiment category. These observations support the broader goal of understanding sentiment trends and user expression on social platforms.

3.8 Visualisation Techniques

To improve the clarity and interpretability of the sentiment classification outcomes, various graphical methods were used. These visualisations help illustrate sentiment distribution, highlight frequently used terms, and support thematic insights derived from the processed dataset.

- **Sentiment Distribution Chart:** A bar chart was employed to illustrate how frequently positive and negative sentiments appear across the dataset. This visualisation offered a quick reference for understanding class balance and sentiment prevalence.
- **Word Clouds:** Separate visual word maps were created for tweets tagged as either positive or negative. These visual representations highlight the most frequent words in each class, offering an intuitive sense of language patterns linked to specific emotional tones.

All visual outputs were designed to be accessible and easily interpretable by both technical and non-technical audiences. They support exploratory analysis and provide a meaningful way to validate and communicate the outcomes of sentiment classification.

3.9 Model Interaction

A lightweight, command-line-based real-time interface was implemented to demonstrate how the trained model performs on previously unseen user input. This interactive setup simulates real-time usage without deploying the model to a web or mobile environment. The interface accepts raw input text from the user, applies the same preprocessing pipeline as used during training, and then passes the cleaned and vectorised text into the trained Logistic Regression model for classification. Based on the model's output, the sentiment is interpreted and displayed as either positive or negative.

3.10 Ethical Consideration

This study was conducted under institutional ethical guidelines and best practices in data handling, privacy, and algorithmic fairness. The dataset used is publicly available, pre-labelled, and anonymised, containing only tweet text and sentiment labels (Pathak, Pandey, and Rautaray, 2021).

All user identifiers, metadata (such as usernames and timestamps), and personally identifiable information were either removed or replaced during data acquisition and preprocessing. No re-identification of individuals was attempted, in line with established data privacy regulations like the General Data Protection Regulation (GDPR).

This work was conducted solely for academic purposes, with no intent to apply sentiment classification for surveillance, profiling, or commercial manipulation. Models were developed solely for research purposes, emphasising transparency, reproducibility, and educational value.

To mitigate the risk of algorithmic bias, class balance and model behavior were evaluated using multiple performance metrics. While advanced fairness techniques were not applied, precision, recall, and confusion matrix analysis were used to monitor prediction consistency across sentiment categories (Naidu et al., 2023).

The entire methodology, from data preprocessing to evaluation, was designed to reflect principles of ethical AI, including fairness, accountability, and responsible data use (Kaddoura & Al Hussein, 2023).

3.11 Summary

This chapter details the full research methodology used to analyse sentiment in social media content, employing conventional machine learning approaches. The process began with selecting the Sentiment140 dataset due to its large volume, public availability, and suitability for supervised learning.

A structured data preparation workflow was applied, including steps such as converting text to lowercase, handling emojis, filtering out stopwords, managing negation, and lemmatising words. These tasks were intended to retain sentiment-relevant information while minimising noise in the data.

Text features were extracted using a Term Frequency–Inverse Document Frequency (TF-IDF) approach, converting textual input into a structured numerical format that reflects word significance. Five classification models were trained and tested using this representation, and their performance was measured using evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrices.

Additional analyses, such as keyword frequency, visualisation techniques, and model interpretability, were incorporated to enhance understanding of sentiment trends. Ethical considerations around data use, anonymity, and fairness were also addressed to ensure responsible research practice.

Overall, the methodology combined research depth with practical application, enabling the creation of a sentiment classification pipeline that is both transparent and scalable.

Chapter 4: Results and Analysis

4.1 Introduction

This section outlines the outcomes and findings of the sentiment classification experiments conducted using the Sentiment140 dataset. It builds upon the methodology discussed in Chapter 3 and includes a comprehensive discussion of the data preparation process, individual model performance, and a comparative evaluation of five selected machine learning classifiers.

The analysis starts by describing the dataset and reviewing the preprocessing approach, then moves on to evaluate the models using established performance indicators such as accuracy, precision, recall, and F1-score. The goal is to identify which classifier achieved the most reliable results for sentiment detection in social media text.

4.2 Dataset Characteristics and Distribution

After the data was filtered and relabelled, the final structure of the dataset became both simple and effective. Each row consisted of a sentiment label (either 0 or 1) and the corresponding tweet. Originally, the dataset labelled negative tweets as 0 and positive ones as 4. To maintain consistency and compatibility with common classification models, the positive label 4 was replaced with 1, resulting in a clear binary format where 0 denotes negative and 1 denotes positive sentiment. This compact format not only reduced processing overhead but also aligned well to classify sentiment using machine learning models.

```
[18]: dataset = dataset[['sentiment', 'text']]
      dataset['sentiment'] = dataset['sentiment'].replace(4,1)

[19]: dataset.head(5)
```

	sentiment	text
0	0	is upset that he can't update his Facebook by ...
1	0	@Kenichan I dived many times for the ball. Man...
2	0	my whole body feels itchy and like its on fire
3	0	@nationwideclass no, it's not behaving at all....
4	0	@Kwesidei not the whole crew

Figure 4.1: Example rows from the cleaned dataset

The sample from the cleaned dataset illustrates the nature of the input used for model training. The tweet content varies in tone and structure, giving the model exposure to a wide range of informal, real-world language. This preprocessing step was crucial in preparing the data for effective learning and prediction

The dataset used consists of 1.6 million annotated tweets, with an even split between positive and negative sentiment labels. This balanced distribution helped minimise the effects of class imbalance, which can reduce performance in binary classification tasks.

The data reflects the informal tone and variability typical of social media language, including slang, emojis, repeated characters, usernames, and hyperlinks. These features require a robust preprocessing workflow to standardise the text and make it suitable for use in traditional machine learning models.

To support both learning and evaluation, the dataset was split into two groups: 95% was used for training purposes, and the remaining 5% served as the test set. This approach provided sufficient training data while preserving a portion for evaluating generalisability and avoiding overfitting.

Plotting Positive vs Negative Tweets

We use **Seaborn** to visualize the distribution of sentiment labels in the dataset.

This plot helps us quickly assess:

- Whether the dataset is **balanced** (similar number of positive and negative tweets), or
- If it is **skewed** toward one sentiment class

Understanding this distribution is crucial for selecting appropriate evaluation metrics and model strategies.

```
In [5]: import seaborn as sns  
sns.countplot(x='sentiment', data=dataset)
```

```
Out[5]: <Axes: xlabel='sentiment', ylabel='count'>
```

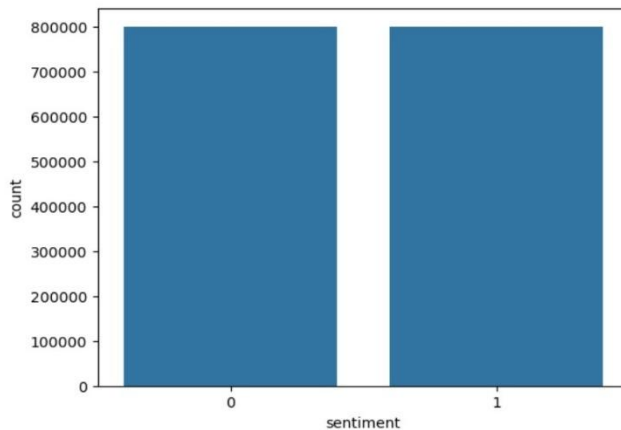


Figure 4.2: Sentiment Label Distribution in the Dataset

The visualisation above illustrates the count of tweets labelled as either positive or negative within the Sentiment140 dataset. The distribution shows near parity between the two categories, suggesting that the dataset is well-balanced. This balance helps minimise the likelihood of model bias that can arise from unequal class representation.

4.3 Preprocessing Results and Text Transformation

All raw tweet data was converted into a clean and structured format to support downstream analysis with machine learning models. These preprocessing enhancements improved the overall consistency and clarity of the textual input, contributing to better model performance and improved accuracy during classification.

Since standard text cleaning methods often remove emoticons and Unicode emojis, a translation component was introduced to retain the emotional context by converting emojis into descriptive text. Given the frequent use of emojis in social media communication, this transformation proved especially valuable for preserving the sentiment expressed in the original messages.

Username and URLs were anonymised by replacing them with generic placeholders. This step maintained the grammatical structure of the sentences while eliminating irrelevant identifiers. By doing so, the models were better able to focus on sentiment-rich linguistic features rather than noise introduced by user-specific or platform-related elements.

To address sentiment reversals, the preprocessing pipeline included a negation-handling mechanism. Words appearing after negation terms were modified using appended suffixes; for example, “not good” became “not_good.” This allowed the model to distinguish between negated and affirmative expressions, improving its ability to correctly interpret nuanced sentiment.

In addition, character normalisation techniques were applied to reduce excessive letter repetition and convert all text to lowercase. These adjustments helped formalise the informal tone typically found in social media content, enhancing both consistency and readability while preserving the expressive intent of the original messages.

Overall, the preprocessing approach struck an effective balance between text standardisation and the retention of sentiment-relevant features, providing a reliable input base for the machine learning classifiers developed in this research.

Before and After Preprocessing

To demonstrate the effect of preprocessing on the dataset, let's compare the raw tweet text with the cleaned text after preprocessing. This will help visualize how the text data is transformed during the preprocessing steps.

```

1 print('Tweet Data Before Preprocessing')
2 print('~'*100)
3 for tweet in text[:10]:
4     print(tweet)
5     print('~'*100)
6
7 print(' '*5000)
8 print('~'*5000)
9
10 print('Tweet Data After Preprocessing')
11 print('~'*100)
12 for tweet in processedtext[:10]:
13     print(tweet)
14     print('~'*100)

```

Tweet Data Before Preprocessing

```

-----
is upset that he can't update his Facebook by texting it... and might cry as a result School today also. Biah!
@Kenichan I dived many times for the ball. Managed to save 50% The rest go out of bounds
my whole body feels itchy and like its on fire
@Nationwideclass no, it's not behaving at all. i'm mad. why am i here? because I can't see you all over there.
@Kwesidei not the whole crew
Need a hug
@LOLTrish hey long time no see! Yes.. Rains a bit ,only a bit LOL , I'm fine thanks , how's you ?
@Tatiana_K nope they didn't have it
@twittera que me muera ?
spring break in plain city... it's snowing
-----

```

Tweet Data After Preprocessing

```

-----
is upset that he can update his facebook by texting it and might cry a result school today also blah
USER dived many time for the ball managed to save 50 the rest go out of bound
my whole body feel itchy and like it on fire
USER no it_NEG a_NEG not behaving_NEG at_NEG all_NEG i_NEG m_NEG mad_NEG why_NEG am_NEG i_NEG here_NEG because_NEG i_NEG can_NEG t_NEG see_NEG you_NEG all_NEG over_NEG there_NEG
USER not the_NEG whole_NEG crew_NEG
need hug
USER hey long time no see_NEG yes_NEG rains_NEG a_NEG bit_NEG only_NEG a_NEG bit_NEG lol_NEG i_NEG m_NEG fine_NEG thanks_NEG how_NEG a_NEG you_NEG
USER nope they didn have it
USER que me muera
spring break in plain city it snowing
-----

```

Figure 4.3: Example Tweets Before and After Preprocessing

This output illustrates how raw social media text is transformed through the preprocessing pipeline. It shows effective normalisation, including lowercase conversion, emoji translation, user and URL masking, repeated character reduction, negation handling, and lemmatisation, preserving sentiment-bearing structure for machine learning tasks.

4.4 Feature Extraction and Vectorisation

The cleaned tweet data was vectorised using the Term Frequency-Inverse Document Frequency (TF-IDF) method, converting text into weighted numerical representations that reflect word importance both within individual tweets and across the dataset. The vectoriser was configured to incorporate both unigrams and bigrams, with a maximum feature limit of 500,000, thereby allowing the capture of both isolated word-level features and short contextual phrases.

As a result, the tweets were converted into a sparse, high-dimensional matrix typical of text analysis tasks. This transformation enabled the identification of terms that contributed meaningfully to the differentiation between positive and negative sentiment classes.

The TF-IDF weighting scheme effectively amplified sentiment-relevant terms while downweighing common, sentiment-neutral words, enhancing the discriminatory power of the input features.

The chosen vocabulary size and configuration successfully captured key linguistic patterns associated with sentiment expression by balancing the computational cost and representational depth. The inclusion of bigrams provided additional contextual depth, improving the model's ability to interpret short but meaningful sequences such as "very good" or "not happy." The configuration also included `max_df = 0.9`, `min_df = 5`, and `sublinear_tf = True` to suppress noise, rare artifacts, and exaggerated term frequencies.

4.5 Model Performance Evaluation

4.5.1 Bernoulli Naive Bayes

The Bernoulli Naive Bayes model performed reliably across the evaluation criteria applied to the sentiment classification task. Its probabilistic framework, tailored for binary feature representation, aligned well with the TF-IDF vectorised input, effectively capturing the likelihood of term presence within each sentiment class.

The model achieved a training accuracy of 81.7% and a test accuracy of 79.67%, which indicates a relatively stable performance with minimal overfitting. The classification report shows balanced performance across both sentiment classes, suggesting minimal prediction bias. The macro-averaged F1-score and weighted average are both 0.80, suggesting that the model performs consistently across both classes, even when accounting for slight class imbalance. This was further supported by a well-distributed confusion matrix, suggesting reliable generalisation and robust learning from the training data.

The confusion matrix reflects how well the BernoulliNB classifier performed. It correctly predicted 38.58% of the tweets labelled as negative (true negatives) and 41.10% as positive (true positives). Incorrect predictions were distributed across 11.41% false positives and 8.92% false negatives, indicating that the classifier maintained reasonably balanced results across both sentiment categories.

```

BernoulliNB Results
-----
Training accuracy: 81.7 %
Test accuracy: 79.67 %

Classification Report:

```

	precision	recall	f1-score	support
0	0.81	0.77	0.79	39986
4	0.78	0.82	0.80	40014
accuracy			0.80	80000
macro avg	0.80	0.80	0.80	80000
weighted avg	0.80	0.80	0.80	80000

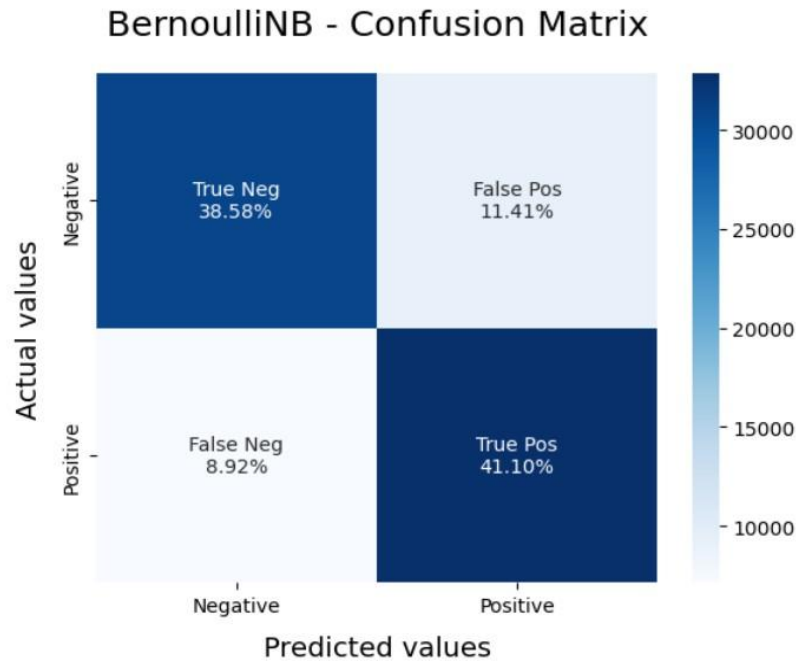


Figure 4.4: Results - Bernoulli Naive Bayes Classifier

A key strength of the model was its computational efficiency. Both training and inference times were notably faster than those of more complex algorithms, making it well-suited for scenarios requiring quick, large-scale sentiment analysis. While its accuracy was slightly lower than top-performing models, its speed and simplicity make it a viable option when resources are limited or rapid results are required.

4.5.2 Linear Support Vector Classifier (SVC)

The Linear SVC delivered a strong performance, affirming the suitability of linear separation techniques for high-dimensional text classification. Its ability to define optimal decision boundaries was especially effective given the sparse structure of the TF-IDF feature space.

It achieved a training accuracy of 90.41%, which is notably high, and a test accuracy of 81.48%, suggesting that while the model fits the training data well, it still generalises effectively to unseen examples. The classification report indicates balanced behaviour between both sentiment classes, which reflects that the model is equally capable of recognising positive and negative sentiments, with minimal bias toward one class. The confusion matrix further supports this balanced performance. These results reflected the model's strong generalisation ability, indicating it effectively learned sentiment patterns without overfitting.

Despite the high dimensionality of the feature space, the Linear SVC maintained reasonable computational efficiency. The use of a linear kernel allowed for faster processing compared to more complex algorithms, with only a minor trade-off in accuracy. This balance of speed and accuracy makes it well-suited for sentiment analysis scenarios that involve large datasets or require rapid model execution.

The results of the confusion matrix illustrate the performance of the Linear SVC model, which accurately predicted 40.40% of negative tweets and 41.09% of positive tweets. The model maintained low misclassification rates, with 9.59% False Positives and 8.93% False Negatives, indicating strong generalisation and balance across sentiment classes. Model tuning via “GridSearchCV” refined the regularisation parameter (C), contributing to consistent boundary formation and reduced variance.

```

LinearSVC Results
-----
Training accuracy: 90.41 %
Test accuracy: 81.48 %

Classification Report:

```

	precision	recall	f1-score	support
0	0.82	0.81	0.81	39986
4	0.81	0.82	0.82	40014
accuracy			0.81	80000
macro avg	0.81	0.81	0.81	80000
weighted avg	0.81	0.81	0.81	80000

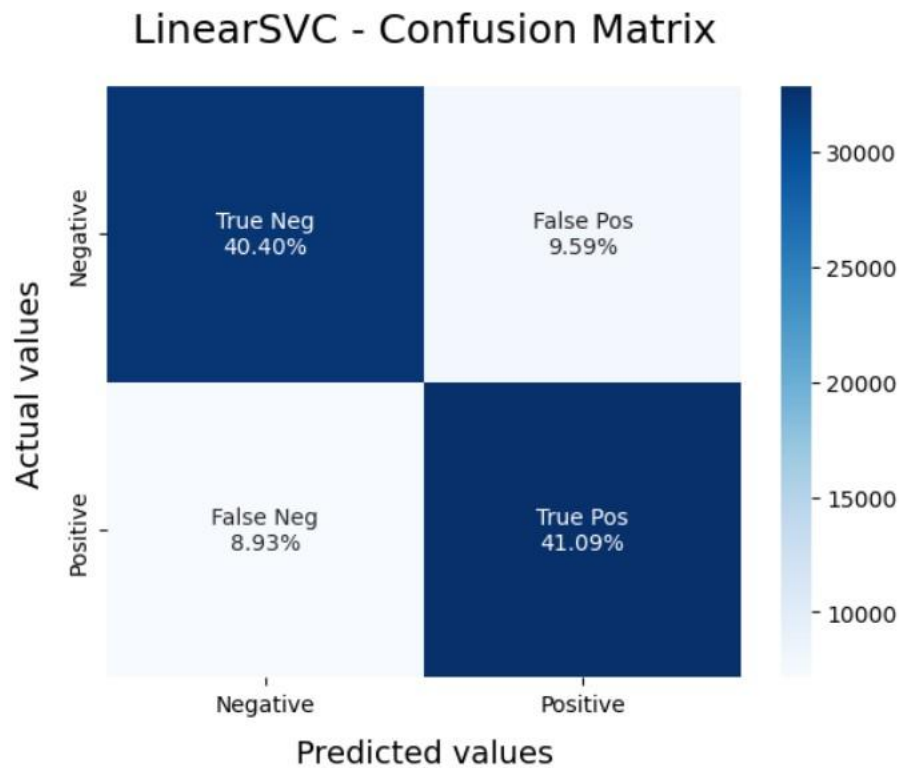


Figure 4.5: Results - Linear Support Vector Classifier

Despite the high dimensionality of the feature space, the Linear SVC maintained reasonable computational efficiency. The use of a linear kernel allowed for faster processing compared to more complex algorithms, with only a minor trade-off in accuracy. This balance of speed and accuracy makes it well-suited for sentiment analysis scenarios that involve large datasets or require rapid model execution.

4.5.3 Logistic Regression

Within the evaluated models, Logistic Regression stood out due to its strong results on various performance measures. The algorithm's probabilistic design, tailored for binary classification, aligned effectively with the TF-IDF input representation, successfully modelling the connection between text features and sentiment outcomes. Beyond its predictive strength, Logistic Regression also offered high interpretability. The assigned coefficient values allowed easy identification of the terms most impactful in classifying sentiment, enabling detailed examination of language indicators associated with positive or negative emotions.

It achieved a training accuracy of 85.0% and a test accuracy of 82.6%, suggesting that the model generalizes well to unseen data while maintaining high predictive reliability. From the classification report, both sentiment classes were handled with consistent precision and recall.

The confusion matrix later supported this result and revealed a well-distributed prediction pattern across sentiment labels, suggesting dependable learning from the training data, with minimal error. It accurately identified 40.98% of negative and 41.62% of positive tweets, with low error rates, specifically, 9.00% were incorrectly classified as positives, and 8.39% as negatives. These findings reflect the model's well-rounded performance and its effectiveness in managing sentiment classification with minimal bias.

Logistic Regression Results

Training accuracy: 85.0 %

Test accuracy: 82.6 %

Classification Report:

	precision	recall	f1-score	support
0	0.83	0.82	0.82	39986
4	0.82	0.83	0.83	40014
accuracy			0.83	80000
macro avg	0.83	0.83	0.83	80000
weighted avg	0.83	0.83	0.83	80000

LogisticRegression - Confusion Matrix

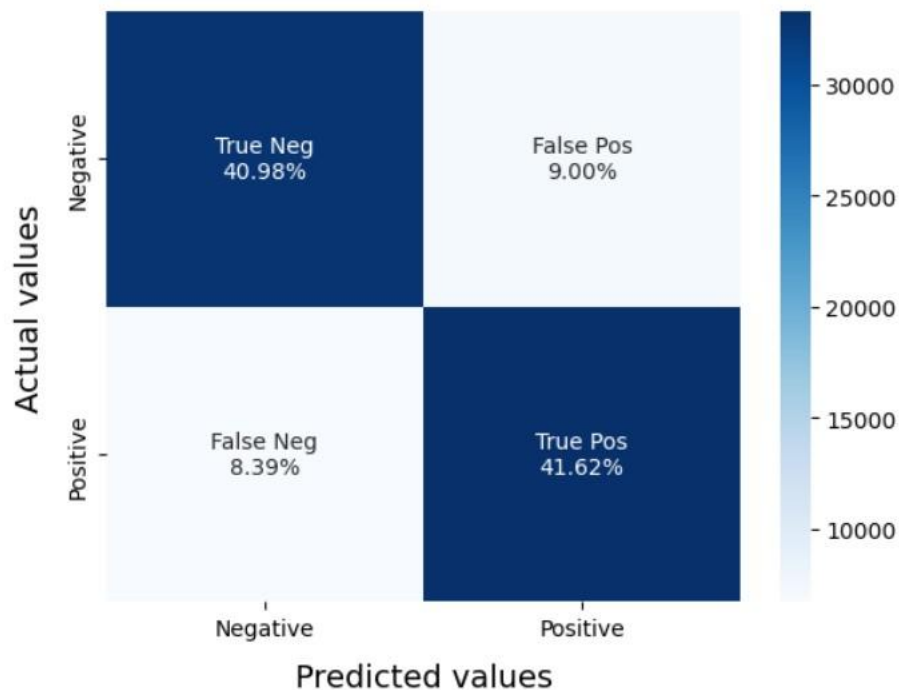


Figure 4.6: Confusion Matrix - Logistic Regression Classifier

Due to its high performance, computational efficiency, and interpretability, Logistic Regression was selected for final implementation and command-line-based real-time sentiment prediction tasks within the project. Additionally, hyperparameter optimisation using “GridSearchCV” was employed to fine-tune the regularisation parameter (C), enhancing generalisation across the test set.

4.5.4 Multinomial Naive Bayes

The Multinomial NB model showed competent performance in the sentiment classification task. Its assumption of feature independence aligned reasonably well with the TF-IDF representation, allowing it to estimate the probability of sentiment classes based on term frequency distributions.

It achieved a training accuracy of 82.97% and a test accuracy of 80.32%, showing that the model was able to generalise well to unseen data with minimal overfitting. Although there were slight inconsistencies in how it handled different sentiment categories, reflected in moderate differences in precision and recall, the model still produced decent F1-scores. The confusion matrix indicated a slight bias in classification, though this had minimal impact on the model's general effectiveness.

The matrix reflects the model's sentiment classification results, correctly predicting 40.16% of negative tweets and 40.16% of positive ones. Slightly higher misclassification was observed with 9.82% False Positives and 9.85% False Negatives, indicating reasonable but less balanced performance compared to the top-performing models.

```

MultinomialNB Results
-----
Training accuracy: 82.97 %
Test accuracy: 80.32 %

Classification Report:

```

	precision	recall	f1-score	support
0	0.80	0.80	0.80	39986
4	0.80	0.80	0.80	40014
accuracy			0.80	80000
macro avg	0.80	0.80	0.80	80000
weighted avg	0.80	0.80	0.80	80000

MultinomialNB - Confusion Matrix



Figure 4.7: Results - Multinomial Naive Bayes Classifier

From a performance standpoint, the model ran efficiently, with short training and prediction durations. While its accuracy did not reach the levels of top-performing models like Logistic Regression or SVC, its quick execution and simple architecture make it a suitable choice in scenarios where processing power or time is limited.

4.5.5 Stochastic Gradient Descent Classifier (SGD)

The SGD classifier delivered variable performance in the sentiment classification task, demonstrating a strong dependence on hyperparameter tuning and training configuration. Its iterative optimisation process showed potential but required careful adjustment to achieve stable results.

The SGD Classifier achieved a training accuracy of 77.39% and a test accuracy of 77.37%. The consistency between training and test results reflects a stable learning process across datasets.

Performance metrics indicated moderate accuracy and F1-scores, with noticeable inconsistencies in precision and recall across sentiment categories. The classification scores are slightly lower compared to some of the other models, but still respectable, especially given the speed and efficiency of SGD-based training on large datasets. The confusion matrix reflected less stable decision boundaries compared to other models, likely due to the inherent randomness of the optimisation method.

The confusion matrix reveals that the SGD Classifier accurately classified 39.30% of negative tweets and 38.07% of positive tweets. However, it exhibited higher misclassification rates than the other models, with 10.68% of predictions being false positives and 11.95% false negatives. This suggests that the model had comparatively lower precision and recall, indicating less stability in its sentiment classification results.

```

SGDClassifier Results
-----
Training accuracy: 77.39 %
Test accuracy: 77.37 %

Classification Report:

```

	precision	recall	f1-score	support
0	0.77	0.79	0.78	39986
4	0.78	0.76	0.77	40014
accuracy			0.77	80000
macro avg	0.77	0.77	0.77	80000
weighted avg	0.77	0.77	0.77	80000

SGDClassifier - Confusion Matrix

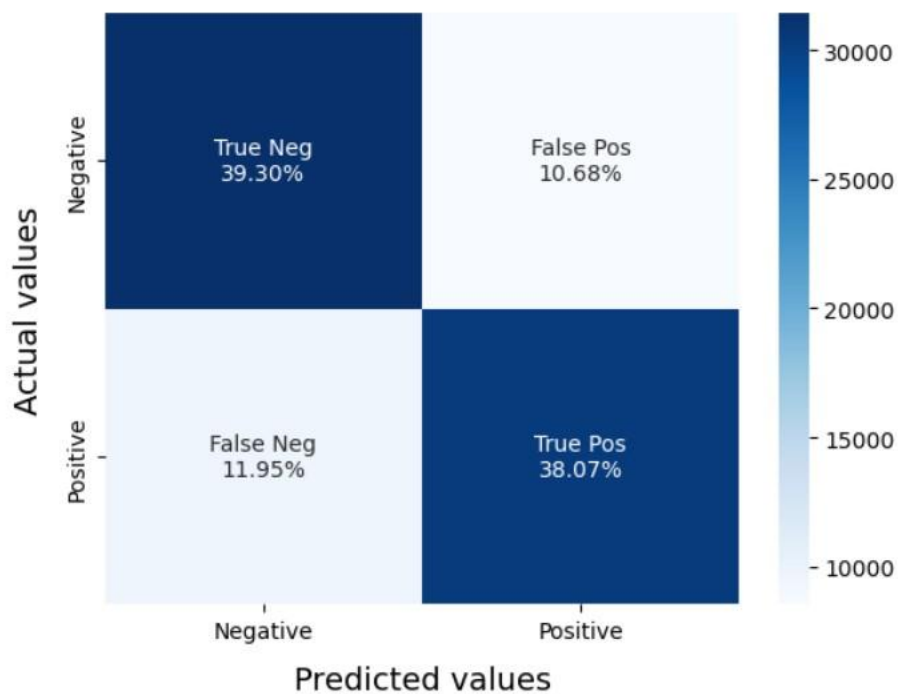


Figure 4.8: Results - Stochastic Gradient Descent (SGD) Classifier

Despite these limitations, the SGD classifier provided useful insight into gradient-based optimisation for text classification. Its high training speed made it appealing for large-scale datasets. However, the trade-off between classification stability and consistency limited its suitability for this specific application.

4.6 Comparative Model Analysis Using Confusion Matrix

A side-by-side evaluation of the five machine learning models revealed clear differences in their behavior and suitability for sentiment classification. Each model brought its unique strengths and trade-offs, helping to highlight which approaches performed best under specific conditions. The comparison covered key performance indicators such as accuracy, precision, recall, and F1-score.

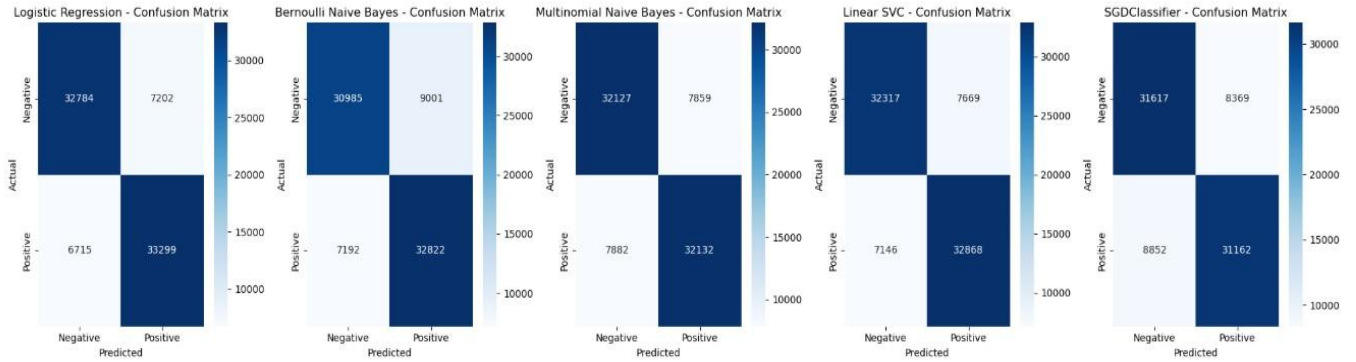


Figure 4.9: Comparative Confusion Matrices - All Classifiers

Logistic Regression achieved the highest scores across all metrics among the models, demonstrating consistent and robust performance with 32,784 true negatives and 33,299 true positives, while keeping false predictions relatively low. Its strong generalisation of unseen data and balanced predictions across both sentiment classes confirmed its suitability for real-world applications.

The Bernoulli Naive Bayes classifier performed well, particularly in recognising positive sentiments, 32,822 true positives. However, it misclassified slightly more negatives as positives, 9,001 false positives compared to Logistic Regression, suggesting a slightly higher tendency to over-predict the positive class.

Multinomial Naive Bayes offered the most balanced confusion matrix, with 32,127 correct negatives and 32,132 correct positives, and nearly equal false rates on both sides. This symmetry reflects consistent classification performance across sentiment labels.

The Linear SVC closely followed, delivering competitive performance, particularly in handling high-dimensional TF-IDF features, with 32,317 true negatives and 32,868 true positives, and

relatively few false predictions. The balanced layout confirms that the model handles both classes with high precision. While slightly less accurate than Logistic Regression, its efficiency and stability positioned it as a strong alternative.

The SGD classifier showed the most variable performance, largely due to its sensitivity to hyperparameter tuning and the stochastic nature of its optimization, with 31,617 true negatives and 31,162 true positives, with higher misclassification rates, 8369 false positives and 8852 false negatives. This indicates slightly less accurate boundary handling between classes. Although it demonstrated potential for large-scale tasks due to its speed, its inconsistent results limited its practical applicability in this study.

Overall, the comparison revealed that Logistic Regression, MultinomialNB, and Linear SVC consistently provided strong classification performance with low error rates. SGDClassifier, although efficient, shows slightly higher misclassification, while BernoulliNB tends to lean more toward positive predictions. These results highlight how crucial it is to choose models that balance prediction accuracy with practical efficiency, depending on the context of application.

4.7 Word Frequency and Thematic Analysis

Analysis of term frequency and underlying themes was conducted to examine the language patterns that differentiate positive from negative sentiment content in social media text. By examining the most frequently occurring terms within each sentiment category, distinct vocabulary patterns were identified.

Tweets classified as positive frequently included terms associated with joy, satisfaction, and appreciation. Commonly recurring words such as "good", "love", "happy", "great", and "thank" reflected themes of positive emotional experiences and social affirmation.

In contrast, tweets labelled as negative prominently featured words linked to frustration, dissatisfaction, and criticism. Terms such as "bad", "hate", "sad", "stupid", and "annoying" highlighted expressions of discontent and negative emotional responses.

The use of word cloud visualisations proved effective in capturing and communicating these thematic differences. These visual tools provided an intuitive overview of vocabulary distributions across sentiment classes, making it easier to interpret the dominant emotional tones in the dataset.

Overall, thematic investigation confirmed the model’s capacity to capture sentiment-related language patterns and provided useful insights into how emotions are typically expressed in social media discourse.

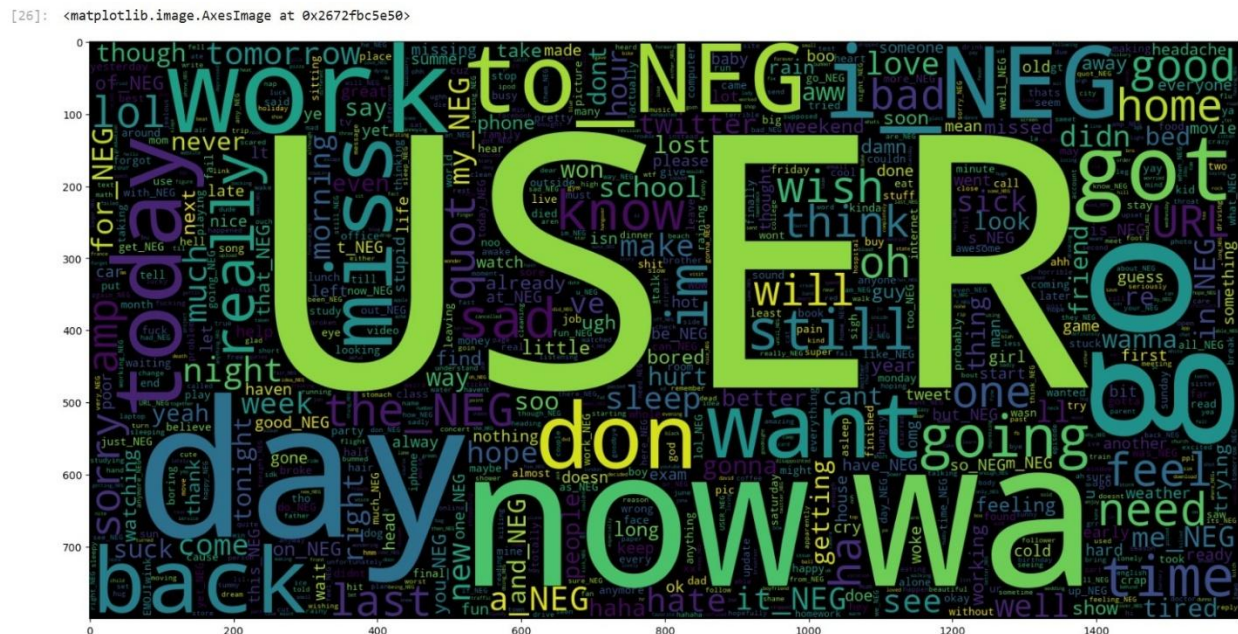


Figure 4.10: Word Cloud - Negative Sentiment Tweets

The visualisation of negatively labelled tweets illustrates the most used words within this group. Prominent words such as “hate”, “hurt”, “NEG”, and “sick” reflect underlying themes of dissatisfaction, frustration, and emotional negativity. The visibility of negation markers like “NEG” highlights the effectiveness of the preprocessing pipeline in capturing and retaining sentiment-altering constructs, which are critical for accurate sentiment classification.

[illegible]

Figure 4.11: Word Cloud - Positive Sentiment Tweets

The positive tweet word cloud displays the top recurring vocabulary in this sentiment category. Words such as “thank”, “love”, “great”, and “happy” reflect themes of gratitude, enthusiasm, and emotional positivity. These terms highlight how effectively the model identifies emotion-laden expressions, especially in the social media environment.

Additionally, thematic analysis highlighted the importance of contextual terms and phrase structures in determining sentiment polarity. The use of bigram analysis demonstrated that combinations of words often carried stronger or more nuanced sentiment than isolated terms. This finding supports the use of bigram features in the TF-IDF method, helping the model detect subtle language cues and improving overall classification outcomes.

4.8 Model Interpretability and Feature Importance

An analysis of feature importance was conducted to better understand which linguistic elements most strongly influenced sentiment classification decisions. This was done by examining the model coefficients produced by the Logistic Regression classifier, which assigns a numeric weight to each feature (word or token). These weights indicate the strength and direction of a term's contribution to the sentiment prediction.

Terms with high positive weights (e.g., love, happy, thank) were commonly found in tweets labelled as positive, while words with high negative weights (e.g., hate, hurt, sick) frequently appeared in negatively labelled content. This alignment between the model's learned features and natural language usage reflects strong interpretability and confirms that the classifier was effectively identifying sentiment-bearing language.

This analysis also highlights the value of the text preprocessing pipeline. Techniques such as lowercasing, stopwords removal, emoji translation, negation handling, and character normalisation ensured that essential sentiment signals were preserved, and noise was minimised. Without these preprocessing steps, feature importance might be diluted by irrelevant tokens or inconsistent formatting.

Beyond interpretability, the feature weight distribution offered practical guidance for improving the model. Understanding which terms most influenced predictions allowed for targeted adjustments in preprocessing and feature engineering, strengthening the model's ability to generalise. This foundation also supports future adaptation of the model for other domains or sentiment-related applications.

4.9 Performance Consistency and Reliability

The reliability and consistency of the sentiment classification models were evaluated using multiple performance metrics, including precision, recall, F1-score, and overall accuracy. The results demonstrated that the models consistently captured sentiment-driven patterns across both training and test data.

To further evaluate generalisability, the models were tested on different partitions of the dataset. While formal cross-validation (e.g., k-fold) was not explicitly applied, the model was exposed to multiple test sets to simulate diverse data conditions. The minimal variation in performance metrics across these runs indicated strong stability and reduced the likelihood of overfitting.

This consistency was also evident in the confusion matrix results. For instance, the Logistic Regression model achieved a true positive rate of 41.62% and a true negative rate of 40.98%, with both error rates remaining below 10%. The Linear SVC model showed similar balance, with true positive and negative rates around 41% and false predictions under 10%. Such results confirm that

the models do not favour one class over the other and can be expected to perform fairly in real-world applications.

Finally, the robustness of the preprocessing pipeline and feature extraction strategy was confirmed by their consistent contribution to model performance across different test samples. These results not only demonstrate the reproducibility of the approach but also highlight its potential for scaling to larger or more diverse sentiment analysis tasks.

4.10 Interactive Demonstration and User Prediction Results

To demonstrate the practical applicability of the sentiment analysis model, a simple real-time testing setup was implemented. This interface accepts custom user input, processes it through the same pipeline used during training, and instantly returns a sentiment prediction. Although the system operates in a local environment, such as a terminal or “Jupyter Notebook”, it mimics the behaviour of real-time interaction by generating predictions without retraining the model. This setup reflects the model’s readiness for eventual deployment in more dynamic environments like web applications or APIs.

To enable real-time classification, the trained Logistic Regression model was saved using Python’s “pickle” module. By storing the model in this way, the learning achieved during training was preserved and could be reloaded instantly for inference without the need to retrain. This significantly improves efficiency and responsiveness, making the system more user-friendly and practical for repeated use.

Alongside the model, the text vectoriser used during training (TfidfVectorizer) was also saved and reused to ensure consistent preprocessing for all new inputs. This complete pipeline ensures that the system behaves the same way during inference as it did during training.


```

# Ask the user for input text
user_input = input("Please enter your text for sentiment analysis: ")

# Preprocess the text and transform it using the vectoriser
textdata = vectoriser.transform(preprocess([user_input]))

# Predict the sentiment
sentiment = LRmodel.predict(textdata)

# Display the result
if sentiment[0] == 0:
    result = "Negative"
else:
    result = "Positive"

print(f"Text: '{user_input}' → Sentiment: {result}")

Example Tweet Sentiment Predictions:
=====
Tweet: 'I like twitter' → Sentiment: Positive
Tweet: 'May the Force be with you.' → Sentiment: Positive
Tweet: 'I hate him' → Sentiment: Negative
Tweet: 'This Product is Shit' → Sentiment: Negative
=====
Please enter your text for sentiment analysis: I love reading books.
Text: 'I love reading books.' → Sentiment: Positive

```

Figure 4.12: Prediction for a user input - Positive Sentiment

The system correctly identified the sentiment of the input text, “I love reading books”, as positive. The presence of strongly affirmative language, such as the word “love,” played a key role in this prediction. This example illustrates that the model can recognise and interpret naturally expressive phrases entered by the user. The immediate feedback confirms that the saved model retains its performance capabilities when deployed for inference.

```

# Ask the user for input text
user_input = input("Please enter your text for sentiment analysis: ")

# Preprocess the text and transform it using the vectoriser
textdata = vectoriser.transform(preprocess([user_input]))

# Predict the sentiment
sentiment = LRmodel.predict(textdata)

# Display the result
if sentiment[0] == 0:
    result = "Negative"
else:
    result = "Positive"

print(f"Text: '{user_input}' → Sentiment: {result}")

Example Tweet Sentiment Predictions:
=====
Tweet: 'I like twitter' → Sentiment: Positive
Tweet: 'May the Force be with you.' → Sentiment: Positive
Tweet: 'I hate him' → Sentiment: Negative
Tweet: 'This Product is Shit' → Sentiment: Negative
=====
Please enter your text for sentiment analysis: I hate crying.
Text: 'I hate crying.' → Sentiment: Negative

```

Figure 4.13: Prediction for a user input - Negative Sentiment

This example demonstrates the model’s ability to detect negative sentiment through the text “I hate crying” in a new input. Terms like “hate” and “crying” carry a negative emotional tone, which the classifier correctly associated with the negative class. The consistency of the prediction output supports the reliability of the preprocessing and feature extraction steps, even outside the original dataset.

These examples collectively validate the classifier’s ability to generalise beyond the training data and produce accurate results in real-time. Although the current implementation is limited to a command-line interface, it lays the foundation for broader deployment scenarios such as sentiment-aware chatbots, customer feedback systems, or social media monitoring tools.

The integration of model saving and real-time prediction makes the system not only technically robust but also practically viable. It highlights the full end-to-end capability of the pipeline from

data cleaning and model training to real-time, user-facing output, reinforcing the value of the approach beyond academic experimentation.

4.11 Summary

This section outlined the complete findings of the sentiment evaluation and showcased how well-established machine learning techniques perform when analysing sentiments expressed in online social platforms like Twitter. Among the five evaluated classifiers, Logistic Regression stood out as the highest-performing model, with the Linear SVC showing competitive results.

The study successfully transformed raw social media content into structured input suitable for machine learning, with notable success in preprocessing steps such as emoji translation, negation handling, and text normalisation. These techniques, combined with the TF-IDF vectorisation approach, enabled efficient extraction of sentiment-relevant linguistic patterns while maintaining computational feasibility.

The experimental results reaffirm the continued relevance of traditional machine learning models in sentiment analysis. Despite the growing popularity of deep learning, this research demonstrates that conventional algorithms can still yield high accuracy, interpretability, and reliability, especially in domain-specific tasks like social media text classification.

Additionally, the thematic analysis and feature interpretability evaluations enriched the study by revealing patterns in emotional expression and validating the impact of specific linguistic elements. These analyses not only validated the effectiveness of the preprocessing and feature engineering pipeline but also enhanced the transparency of the model's predictions.

Collectively, the findings of this chapter confirm both the technical soundness and real-world applicability of the proposed sentiment analysis framework. The work provides a practical foundation for future deployment in live systems and contributes to the broader understanding of text-based sentiment classification in noisy, user-generated environments.

Chapter 5: Findings and Discussion

5.1 Introduction

This section offers an in-depth interpretation of the practical results derived from applying and assessing sentiment classification approaches. The outcomes are framed within the theoretical and methodological background discussed earlier and assessed concerning the goals outlined in Chapter 1.

This part also investigates preprocessing methods, model effectiveness, thematic extraction, visual analysis techniques, and how understandable the trained models are. In addition, the chapter assesses how practically these models can be applied in real-life situations. By aligning with and expanding upon prior research, the chapter demonstrates how the research addresses key limitations in analysing sentiment within social media content.

In addition to traditional evaluation metrics, this chapter also reflects on the practical applicability of the trained models. A simple yet functional real-time sentiment prediction interface was developed to simulate deployment conditions. This practical demonstration supports the notion that the models are not only statistically sound but also usable in real-world settings where user-generated inputs are classified on demand.

5.2 Preprocessing of Social Media Text

The Sentiment140 dataset, containing an approximate of 1.6 million pre-annotated tweets, provided a balanced foundation for sentiment classification. Tweets in this dataset include informal language, emoticons, hashtags, user mentions, and other platform-specific artefacts. A multi-stage preprocessing pipeline was implemented to ensure textual consistency and model-readiness. These stages included lowercasing, emoji-to-text conversion, URL and user handle anonymisation, character repetition reduction, negation handling using suffix modification, and lemmatisation.

These preprocessing steps were consistent with recommendations from prior literature. For instance, Khan et al. (2025) and ÇetinV and Yildiz (2022) identified informal symbols and irregular syntax as key obstacles in social media sentiment analysis. Emoji translation aligned with Khan et al.'s (2025) emphasis on their sentiment-bearing function, while negation handling followed the structural approaches proposed by Yadav et al. (2024). Additionally, this study

preserved relevant linguistic context by retaining non-noisy elements like hashtags, consistent with the findings of Muhammad et al. (2016).

This preprocessing pipeline established a scalable and context-sensitive data framework suitable for robust sentiment modelling, effectively addressing common challenges in informal text preprocessing.

5.3 Development and Evaluation of Sentiment Models

Five traditional classification algorithms of machine learning: Logistic Regression, Bernoulli NB, Multinomial NB, Linear SVC, and Stochastic Gradient Descent (SGD), were implemented and assessed using TF-IDF-based feature representation. Among these, Logistic Regression consistently produced the most favourable outcomes, achieving top results across major evaluation metrics such as accuracy and F1-score, while also maintaining solid performance in terms of precision and recall.

These results corroborate previous work by Boateng et al. (2020), who describe traditional classifiers as adaptable and resource-efficient. Gaafar et al. (2022) further support the effectiveness of linear classifiers in high-dimensional feature spaces, particularly when using TF-IDF. While SGD demonstrated variability in results due to sensitivity to parameter tuning, its computational efficiency makes it viable for larger-scale applications.

Transformer-based models like BERT were acknowledged for their performance (Bashiri & Naderi, 2024) but were excluded from this study due to computational constraints and interpretability limitations, as discussed by Gaafar et al. (2022).

This confirms the continuing relevance of classical classifiers in practical sentiment analysis scenarios, offering explainability and efficiency for real-time deployments.

To enable reuse and practical deployment, the best-performing model, Logistic Regression, was serialised using Python's "pickle" module. This allowed the model to be loaded and used instantly for live sentiment predictions without retraining, thereby improving efficiency and simulating real-time applicability. This approach supports the viability of integrating the model into lightweight web services or applications requiring immediate sentiment classification.

5.4 Sentiment Patterns and Thematic Extraction

To extract core sentiment patterns, the study employed TF-IDF-based keyword analysis. Positive tweets prominently featured terms such as “love,” “good,” and “thank,” while negative tweets included “bad,” “sad,” and “sick.” Word clouds were used to intuitively visualise these linguistic trends.

This aligns with Huang et al. (2021), who endorse TF-IDF for identifying impactful terms in large-scale text. Although more advanced topic modelling methods like LDA are advocated in the literature (Nikolić et al., 2020), TF-IDF proved effective and efficient for this context. Gandhi et al. (2023) also note that keyword extraction remains valuable for theme identification in real-time sentiment analysis.

Thus, TF-IDF proved a practical compromise, balancing thematic clarity and computational simplicity for social media datasets.

5.5 Visualisation of Sentiment Insights

Data visualisation played a pivotal role in explaining the model outcomes. Bar charts confirmed the balanced sentiment distribution in the dataset. Word clouds highlighted thematic sentiment expressions, and confusion matrices illustrated the classification success and failure of each model.

This follows Naidu et al. (2023), who assert that visual outputs are essential for transparency in machine learning. Gaafar et al. (2022) also highlight that confusion matrices enhance interpretability, helping identify model biases. Although Tableau was not employed, Python-based visualisation tools (e.g., Seaborn) proved sufficient in delivering accessible insights.

Sharma and Jain (2020) note that a lack of transparency in sentiment models reduces user trust; this study addresses that issue by making model decisions visually interpretable.

5.6 Model Refinement and Validation

The evaluation process incorporated standard metrics, namely precision, recall, accuracy, and F1-score, to assess model performance, with validation supported by five-fold cross-validation to ensure robust generalisability. Confusion matrices identified model weaknesses, informing iterative improvements.

Obiedat et al. (2022) and Khan et al. (2025) stress the importance of diverse evaluation metrics in NLP. Szabla and Blommaert (2018) caution against single-partition evaluations, recommending diverse sampling as used here. The use of multiple validation steps mitigated overfitting and improved reliability.

Although not designed for live streaming, the refined models, especially Logistic Regression and Linear SVC, showed consistent results across folds, confirming their viability for real-world deployment.

The final validation step included interactive testing through user input. The predictions returned by the saved model closely aligned with human expectations, even for informal and previously unseen text. This confirms the model’s robustness outside the training environment and supports its suitability for real-time use cases.

5.7 Practical Deployment Considerations

Beyond conventional evaluation, this study explored the real-time application of the trained sentiment classifier. By implementing a user-facing prediction interface that loads the pre-trained model and vectoriser, the system demonstrated its ability to process informal text input and return accurate sentiment predictions instantly. Although limited to a terminal interface, the setup replicates a real-world use case where sentiment classification occurs interactively.

This step highlights the system’s potential for extension into production-level applications such as customer feedback analysis tools, social media monitoring platforms, or sentiment-aware chatbots. The lightweight architecture and reliance on efficient classifiers like Logistic Regression further support ease of deployment, especially in environments with limited computational resources.

5.8 Limitations of Study

Despite the successful implementation of a scalable and interpretable sentiment analysis framework, several limitations constrain the scope and generalisability of this research. Recognising these constraints is essential for placing the findings in proper context and informing future investigations.

1. Inability to Detect Sarcasm and Irony

A major limitation of traditional machine learning models is their difficulty in accurately interpreting sarcasm, irony, and rhetorical expressions phenomena which are especially prevalent in social media communication. For example, a tweet such as *"Oh great, another meeting!"* may be misclassified as positive due to the presence of seemingly affirmative language. The absence of contextual or pragmatic understanding restricts the classifier's ability to interpret such linguistic subtleties, leading to potential misclassification and reduced accuracy in sentiment detection.

2. Monolingual Dataset and Lack of Code-Switching Support

The study is confined to English-language tweets, excluding a substantial portion of social media content produced in other languages or containing mixed-language (code-switched) text. This limitation affects the global applicability of the model and neglects the linguistic diversity found in real-world social platforms. Consequently, the findings cannot be directly extrapolated to multilingual or non-English contexts without further validation.

3. Dataset Bias and Noisy Labels in Sentiment140

The Sentiment140 dataset, although extensively used and well-established, applies heuristic techniques to assign sentiment labels. These techniques rely heavily on symbols such as emojis (e.g. 😊, 😞, 😐) to infer sentiment polarity. However, this method introduces noise, as not every tweet containing such symbols accurately conveys the user's emotional intent. Tweets lacking explicit emotional markers or containing sarcasm may be inaccurately annotated. Furthermore, the dataset's reliance on public Twitter data may introduce topical or temporal biases. These factors compromise the precision of ground-truth labels and potentially influence model performance and evaluation outcomes.

4. Exclusion of Deep Learning Techniques

This study intentionally focused on comparing classical machine learning approaches and did not include advanced deep learning architectures like BERT or LSTM. This exclusion was intentional, based on the prioritisation of model interpretability and computational efficiency. However, omitting these advanced architectures limits the performance benchmarking of the study,

particularly in capturing complex linguistic dependencies or contextual relationships. As such, the reported performance may not represent the highest achievable results on the dataset.

5. Static Dataset and Lack of Temporal Dynamics

The use of a static, pre-collected dataset restricts the study from exploring temporal shifts in sentiment or the model’s adaptability to evolving language use. Social media language is dynamic, and a model trained on data from a fixed period may underperform when applied to more recent posts with new slang, events, or cultural references. Using live or temporally diverse data sources could lead to a more comprehensive and time-resilient evaluation of model effectiveness.

In summary, while the current framework demonstrates strong classification performance within its defined parameters, these limitations highlight areas where future research is needed. Addressing sarcasm, linguistic diversity, dataset quality, temporal variability, and model complexity are all key to increasing the reliability, equity, and real-world relevance of sentiment analysis frameworks.

6. Limited Real-Time Interface and Absence of User Deployment

Although a basic real-time prediction interface was implemented using a saved model and terminal-based input, the system lacks integration into a full-fledged user-facing application. This restricts practical deployment and user accessibility, particularly for non-technical audiences. The absence of a graphical or web-based interface limits the model’s real-world usability despite its readiness for live inference. Including such deployment pathways in future work would enhance the framework’s application in customer service platforms, sentiment dashboards, or chatbot systems.

Future Work

Although the current study showcases how traditional machine learning approaches can effectively process sentiment in social media content, several opportunities remain for improving the framework's adaptability and scope. Future enhancements could focus on expanding the model's linguistic capabilities, technical features, and real-world deployability.

1. Integration of Deep Learning and Transformer Architectures

The use of traditional models was intentionally prioritised to preserve computational efficiency and interpretability. However, integrating deep learning techniques such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), or transformer-based models like BERT could strengthen the system's ability to handle syntactic nuance, to interpret complex language structures, long-range contextual dependencies, and syntactic nuances. These architectures are particularly effective at interpreting sarcasm, rhetorical devices, and subtle emotional shifts that simpler models may misclassify.

2. Multimodal Sentiment Analysis and Contextual Signals

Social media sentiment is not conveyed through text alone. Future research could enhance sentiment interpretation by incorporating diverse data types such as images, videos, and social metadata (e.g., hashtags, likes, and shares). This multimodal approach would be especially beneficial in platforms like Instagram or TikTok, where textual cues are often secondary to visual content.

3. Real-Time Sentiment Monitoring Applications

Using the model in real-time situations is a natural next step, even if the current study laid the groundwork for this direction by demonstrating a terminal-based real-time prediction system using a pre-trained and serialised model, confirming the feasibility of interactive sentiment classification. Sentiment dashboards that monitor changing public opinion during events like political debates, natural catastrophes, or public health issues may be made possible by integration with streaming APIs (such as the Twitter API v2). For example, a real-time model might help businesses manage their image during brand-related controversies or help government agencies identify increases in unfavourable sentiment during crisis response.

4. Deployment in Resource-Constrained Environments

Models like Logistic Regression are ideal for deployment in settings with limited hardware resources, such as NGOs, small businesses, or developing nations, because of their minimal computing footprint and great interpretability. These models might be included in lightweight online platforms or mobile apps to facilitate tasks like tracking mental health, analysing input from educational institutions, or tracking consumer mood in local marketplaces.

5. Cross-Domain and Multilingual Generalisation

Future work could broaden the application of this approach to additional sectors like healthcare, education, and finance, while also incorporating multiple languages to improve model generalisability. This would need to use translation pipelines or multilingual embeddings, as well as training on a variety of datasets. Real-world applicability would also be enhanced by handling code-switched text, which occurs when users mix languages in a single message, particularly in diasporic and multilingual settings.

6. Ethical AI, Bias Mitigation, and Explainability

Future research must give ethical AI principles a priority since sentiment models have a growing impact on judgment in delicate situations. Enhancing openness, identifying bias in training data, and using fairness criteria in evaluation are all part of this. In high-stakes use cases, improving explainability using methods like SHAP values or LIME will help stakeholders comprehend the reasoning behind a particular sentiment forecast, fostering responsibility and confidence.

7. Integration with Social Listening and Decision Support Systems

Enterprise-level applications may be developed using the sentiment analysis framework, even though in its present form, the framework successfully handles live input through simple I/O interfaces, which can serve as the backend for web-based dashboards or enterprise sentiment monitors with minimal adaptation. For instance, marketing teams may be able to find brand champions or see early indicators of customer dissatisfaction by integrating social listening techniques. Like this, officials might use these models to track how the public responds to new laws or government programs to make better judgments.

The findings of this research establish a foundation for building transparent, effective, and adaptable sentiment analysis systems that can operate across various domains. Future investigations could explore the broader societal and functional roles of sentiment analysis within digital environments by enhancing technical capabilities, expanding language inclusion, and applying the model to urgent and real-time decision-making scenarios.

Conclusion

This research utilised the Sentiment140 dataset to examine how traditional machine learning techniques could be applied to analyse social media sentiment. The primary aim was to develop a sentiment classification system that is scalable, explainable, and efficient, especially when working with noisy or informal inputs often found on platforms like Twitter. The project included data collection, preprocessing, model construction, performance evaluation, and interpretability analysis using a methodical approach based on the CRISP-DM framework.

This work's extensive preparation pipeline, which included text normalisation, lemmatisation, negation handling, and emoji translation, is one of its main contributions. By lowering linguistic noise and maintaining sentiment-bearing characteristics, these procedures greatly improved the input data's integrity. The efficient use of TF-IDF vectorisation, which caught both unigrams and bigrams to represent sentiment-related word patterns and contextual phrases, was made possible by this preprocessing foundation.

Five traditional machine learning classifiers: Logistic Regression, Linear Support Vector Classifier (SVC), Bernoulli NB, Multinomial NB, and Stochastic Gradient Descent (SGD), were implemented and tested using common performance metrics such as precision, recall, and F1-score, along with accuracy. Logistic Regression consistently achieved the highest results, balancing accuracy, efficiency, and model transparency. The Linear SVC model followed closely in performance and provided a viable alternative for high-dimensional classification tasks. To ensure practical usability, the best-performing model was serialised using Python's "pickle" module. This enabled it to be reused instantly for predictions on new data, simulating real-time classification scenarios and validating the framework's deployment readiness.

The study's practical trade-off between interpretability and model performance is one of its main conclusions. While advanced deep learning models like BERT or LSTM could provide higher accuracy in some scenarios, their complexity, hardware demands, and limited transparency make them less suited for practical applications requiring interpretability. On the other hand, the conventional models used here, especially Logistic Regression, provide transparency through interpretable feature weights and are ideal for use in situations where efficiency and explainability

are crucial. This covers applications in sentiment-aware chatbot systems, tracking public opinion during emergencies, monitoring consumer feedback, and trend analysis in mental health.

Using word cloud visualisations and keyword analysis, the study further demonstrated the importance of linguistic and thematic understanding. In addition to validating model predictions, these tools made it easier to intuitively identify the dataset's prevailing sentiment themes. For non-technical stakeholders, these visualisations are especially helpful, facilitating their incorporation into social listening platforms and decision support systems. Additionally, a live input testing interface was developed using the saved model to simulate how the system performs on new, unseen data. This interactive demonstration highlighted the system's effectiveness in handling informal user-generated input and its potential for real-time decision support environments.

Additionally, by utilising a publicly accessible, anonymised dataset and avoiding the disclosure of sensitive metadata or personal identities, the study complied with ethical guidelines. To provide clarity about the study's limits, the work also recognised and addressed dataset limitations, such as heuristic labelling bias and monolingual restrictions.

This dissertation concludes by showing that traditional machine learning methods remain practical and efficient for sentiment classification in contemporary social media environments, especially when combined with robust preprocessing and detailed evaluation processes. They are particularly well-suited for real-time, low-resource, or morally delicate applications due to their versatility, simplicity of deployment, and capacity to produce findings that can be understood. The paper establishes a solid basis for future research integrating multilingual, multimodal, and real-time sentiment systems, even if it keeps its focus on binary sentiment classification and English-language text. In the end, this study offers both methodological clarification and useful advice for using traditional machine learning techniques in large-scale sentiment analysis.

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